

- 1) a) V b) F c) V d) V e) V

2)

8400	2
4200	2
2100	2
1050	2
525	3
175	5
35	5
7	7
1	

$$8400 = 2^4 \cdot 3^1 \cdot 5^2 \cdot 7^1$$

3) a) $6^4 \cdot \frac{1}{3^4} = \frac{2^4 \cdot 3^4}{3^4} = 2^4 = 16 \quad V$

b) $7^0 = 1 \neq 0 \quad F$

c) $0^0 = \frac{7^1}{7^1} = 1$

$(-2)^4 = (-2)(-2)(-2)(-2)$

$= 4 \cdot 4 = +16 \neq -16 \quad F$

d) $5^2 \cdot 5^3 = 5^{2+3} = 5^5 \neq 5^6 \quad F$

e) $2^{-3} = \left(\frac{1}{2}\right)^3 \quad V$

f) $11^{-1} = \frac{1}{11} \neq -11 \quad F$

4) DEVO TROVARE L' m. c. m di

54	2
27	3
9	3
3	3
1	

18	2
9	3
3	3
1	

81	3
27	3
9	3
3	3
1	

$18 = 2^1 \cdot 3^2$

$54 = 2^1 \cdot 3^3$

$81 = 2^0 \cdot 3^4$

$m.c.m = 2^1 \cdot 3^4$

$= 162$

5)

$$a) 0,32 = \frac{32}{100} = \frac{16}{50} = \frac{8}{25}$$

$$b) x = 0,3\overline{1}$$

$$10x = 3,1\overline{1}$$

$$10x - x = 3,1\overline{1} - 0,3\overline{1}$$

$$9x = \frac{2,8}{9}$$

$$x = \frac{28}{90} = \frac{14}{45}$$

c)

$$x = 2,4\overline{2}$$

$$100x = 242,4\overline{2}$$

$$100x - x = 242,4\overline{2} - 2,4\overline{2}$$

$$99x = 240$$

$$x = \frac{240}{99}$$

6)

$$\left[1 + (2^6 : 2^3)^0 \right]^2 : 2^2 \cdot \left[-2 + (-3)^5 : (-3)^3 - (-3) \right] \cdot 5^{-1}$$

$$= \left[1 + (2^3)^0 \right]^2 : 2^2 \cdot \left[-2 + (-3)^2 + 3 \right] \cdot 5^{-1}$$

$$= \left[1 + 1 \right]^2 : 2^2 \cdot \left[-2 + 3^2 + 3 \right] \cdot 5^{-1}$$

$$= \frac{2^2}{2^2} \cdot \frac{10^2}{5} = 2$$

7)

$$\left\{ \frac{3}{10} + \left(-\frac{6}{5} \right) \cdot \left[-1 - \frac{5}{6} \right] \right\} : \left[-\frac{1}{4} + \left(-2 + \frac{5}{3} \right) \cdot \frac{9}{4} \right]$$

$$= \left(\frac{3}{10} - \frac{6}{5} \cdot \left(-\frac{11}{6} \right) \right) : \left(-\frac{1}{4} + \left(\frac{-6+5}{3} \right) \cdot \frac{9}{4} \right)$$

$$= \left(\frac{3}{10} + \frac{11}{5} \right) : \left(-\frac{1}{4} - \frac{1}{3} \cdot \frac{3^2}{2^2} \right)$$

$$= \frac{3+22}{10} : \frac{-1}{12} \cdot (-1)$$

$$= -\frac{25}{10} = -\frac{5}{2}$$

$$8) \left\{ \left[\left(\frac{125}{27} \right)^2 \cdot \left(\frac{4}{25} \right)^3 \right]^4 : \left[\frac{8}{27} \right]^5 \right\} : \left(\frac{2}{3} \right)^{10}$$

$$= \left(\left(\left(\frac{5^3}{3^3} \right)^2 \cdot \left(\frac{2^2}{5^2} \right)^3 \right)^4 : \left(\frac{2^3}{3^3} \right)^5 \right) : \left(\frac{2}{3} \right)^{10}$$

$$= \left(\left(\frac{\cancel{5^6}}{3^6} \cdot \frac{2^6}{\cancel{5^6}} \right)^4 \cdot \frac{3^{15}}{2^{15}} \right) \cdot \frac{3^{10}}{2^{10}}$$

$$= \left(\frac{2^{24}}{3^{24}} \cdot \frac{3^{15}}{2^{15}} \right) \cdot \frac{3^{10}}{2^{10}}$$

$$= \frac{2^{24-15}}{3^{24-15}} \cdot \frac{3^{10}}{2^{10}} = \frac{2^9}{3^9} \cdot \frac{3^{10}}{2^{10}} = \frac{3}{2}$$